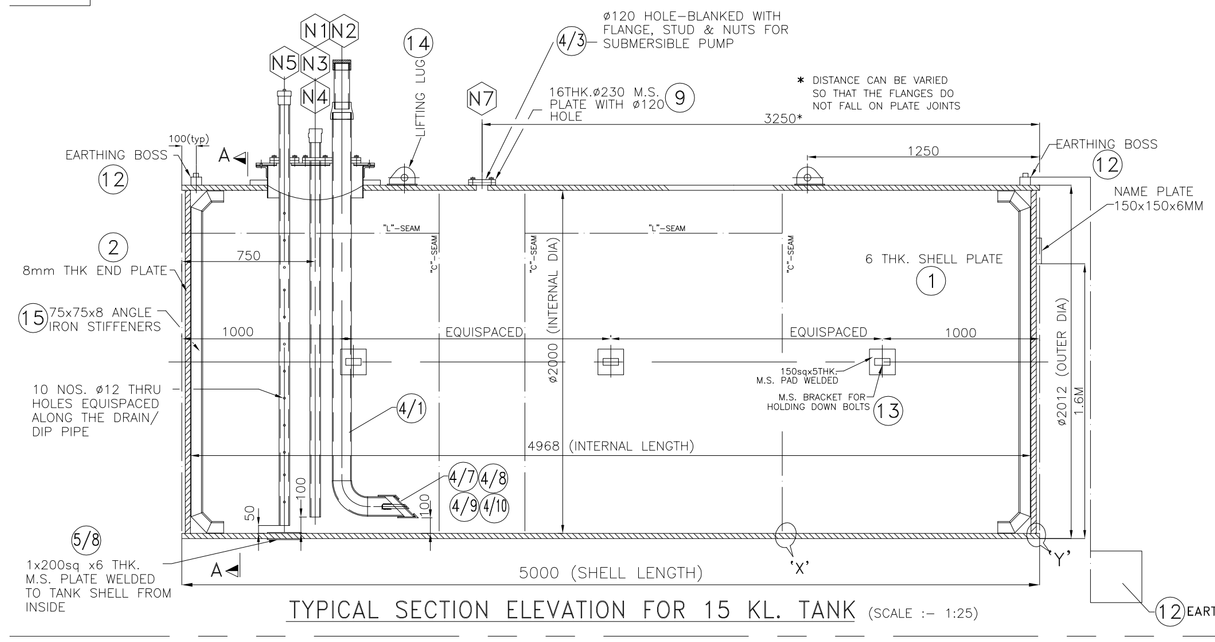
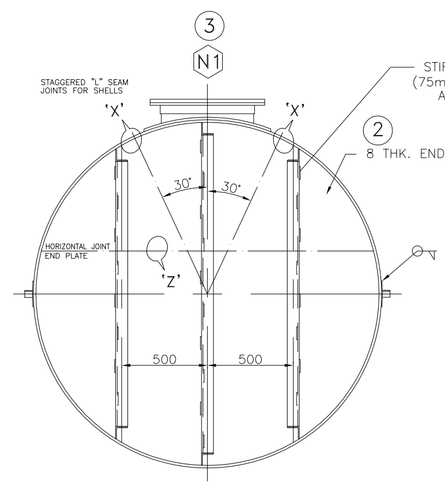
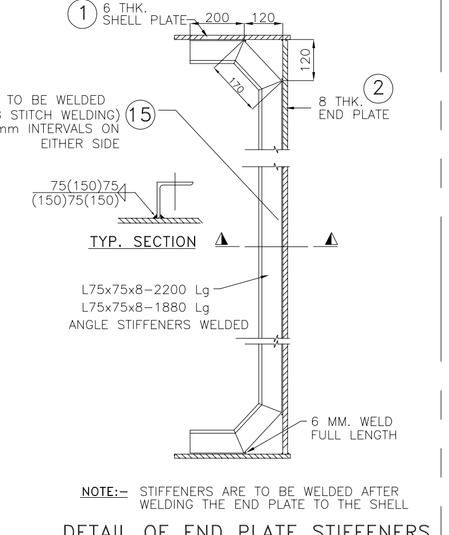
RE. DRG. 002



TYPICAL SECTION ELEVATION FOR 15 KL. TANK (SCALE :- 1:25)

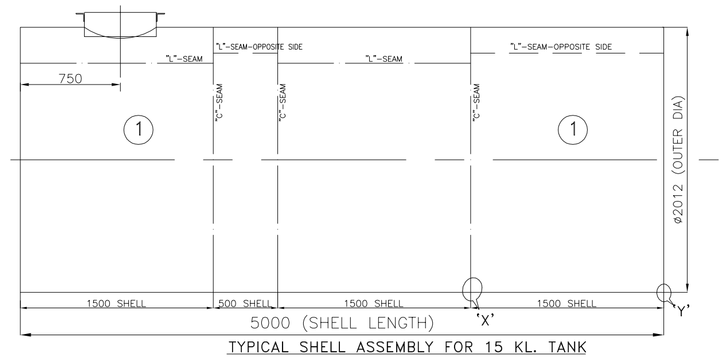


SECTION A–A

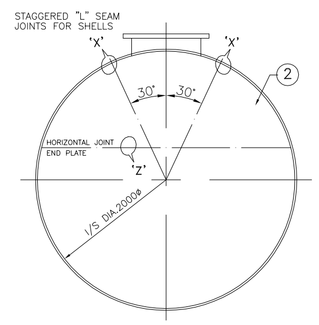


NOTE:– STIFFENERS ARE TO BE WELDED AFTER WELDING THE END PLATE TO THE SHELL

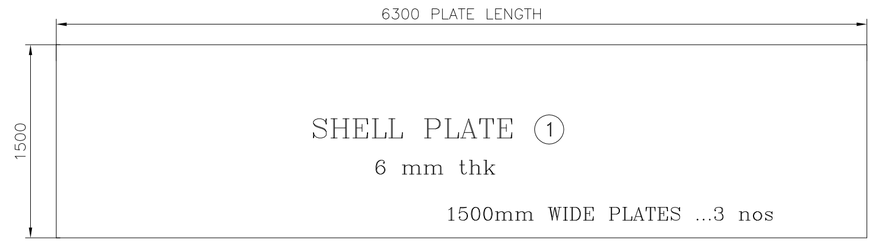
DETAIL OF END PLATE STIFFENERS
ENLARGED VIEW

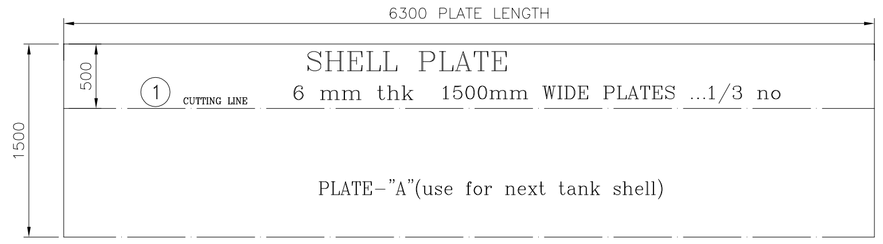


TYPICAL SHELL ASSEMBLY FOR 15 KL. TANK

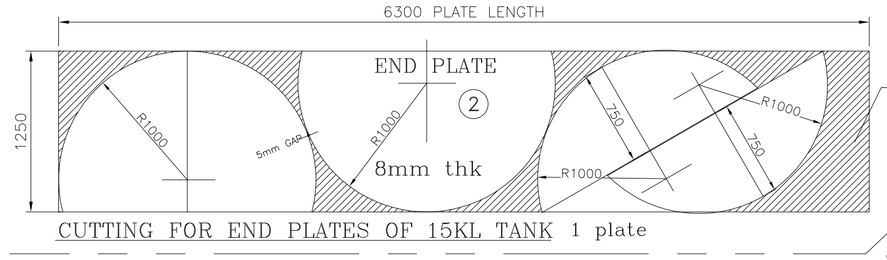


TYP.SECTION

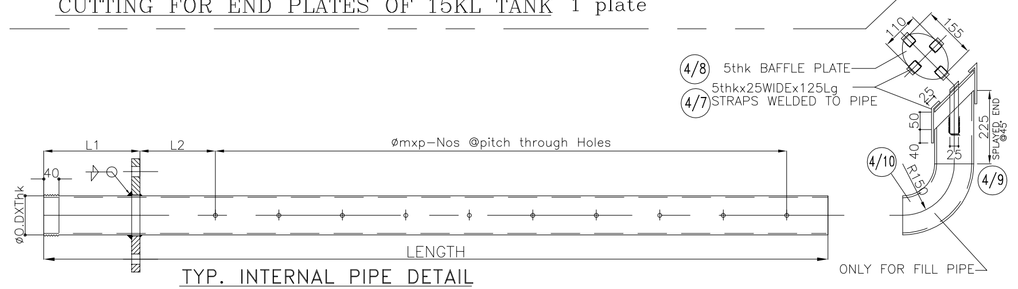




CUTTING FOR SHELL PLATE OF 15KL TANK



CUTTING FOR END PLATES OF 15KL TANK 1 plate



TYP. INTERNAL PIPE DETAIL

TOLERANCE

TOLERANCE ON SHELL DIAMETER – THE TOLERANCE ON SHELL DIAMETER AT ANY POINT ON THE LENGTH OF THE TANK SHELL SHALL BE ±0.35% OF THE SPECIFIED DIAMETER (MAX) WHEN CALCULATED FROM AN EXTERNAL CIRCUMEFERENTIAL MEASUREMENT.
TOLERANCE OF CIRCULARITY – THE TOLERANCE ON CIRCULARITY AT ANY POINT ON THE LENGTH OF THE TANK SHELL THAT IS THE DIFFERENCE BETWEEN MAXIMUM AND MINIMUM INTERNAL SHELL DIAMETER SHALL NOT EXCEED 1% OF NOMINAL INTERNAL DIAMETER.
TOLERANCE ON OVERALL LENGTH – TOLERANCE ON OVERALL LENGTH OF THE TANK SHALL NOT DIFFER BY MORE THAN 0.5% OF THE OVRALL LENGTH.

CUTTING SUMMARY FOR 1 No. 15KL TANK

8mm PLATES FOR END PLATES(1250mm)	..... 1p	495 kg
6mm PLATES FOR SHELL (1500mm PLT)	... 3.33p	1483 kg
NETT WEIGHT OF PLATES USED FOR 1 No. TANK ...		1978 kg

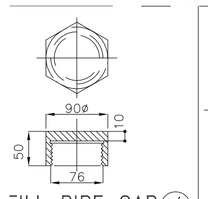
CUT PIECES SCHEDULE

A	6mmX1000X6300....1 no. FROM 1500mm WIDE PLATE TO BE USED FOR NEXT 15KL SHELL
B	8mm CUT PIECES AS SHOWN HATCHED, TO BE USED FOR DATUM, SPLASH PLATES, etc.

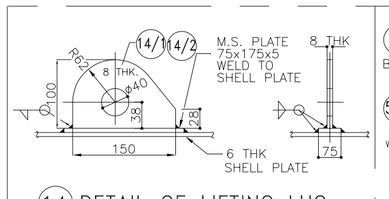
NOZZLE/PIPE/FLANGE SCHEDULE

N7		100NB	BLANK CONN. (SUBMERSIBLE PUMP)							9	230	16	19.0x8	190	WITH BLIND FLANGE
N6	11	50NB	VENT PIPE CONNECTION	60.3	4.5										
N5	5	50NB	DIP / DRAIN PIPE ASSEMBLY FOR DRAWING OUT WATER REMOVE DIP–ROD & INSERT ø25 TUBING	60.3	4.5	2250	155	235	12x10@200	5/3	150	16	19.0x4	120	
N4	7	50NB	SUCTION PIPE ON TANK SHELL WITH 50NB ANGLE CHECK VALVE REFER DRG.No.CD/R–23	60.3	4.5	2050	125	--	---	5/3	150	16	19.0x4	120	
N3	6	100NB	BLIND FLANGE CONNECTION FOR ATG	114.3	5.4	2130	280	--	---	5/3	150	16	19.0x4	120	
N2	4	100NB	TANK LORRY DISCHARGE/FILL PIPE	114.3	5.4	2130	280	--	---	4/3	230	16	19.0x8	190	BOTTOM WITH BEND & CUT @45° BAFFLED
N1	3	ID520ø	MANHOLE COVER ASSEMBLY							3/1	662	10	17.5x24	612	MANHOLE COVER
Nozzle No.	Sr. No.	SIZE NB	NOZZLE CONNECTION	øO.D	Thk	LENGTH	L1	L2	øMxP–Nos@pitch	Sr. No.	øD	thk	øDxn–Nos	øpcd	REMARKS
				PIPE DETAILS							FLANGE DETAILS				

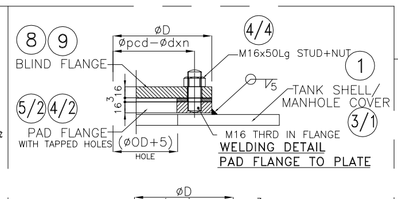
NOS.	LIST OF FITTINGS – DESCRIPTION	MATERIAL	QTY.	SIZE	WT(kg) 2411.28
17	REDUCER (NOT IN FABRICATORS SCOPE OF SUPPLY)	M.S.–IS1239	1	80NBX100NB–"C" CLASS	--
16	FILL PIPE SPOOL–80NB (NOT IN FABRICATORS SCOPE OF SUPPLY)	M.S.–IS1239	1	ø88.9odx4.8thk	--
15	STIFFENERS WELDED TO TANK END PLATES.	M.S.–IS2062	2	L75X75X8–6000Lg(total)	126.96
14/2	REINFORCEMENT PAD PLATE	M.S.–IS2062	1	75x175x5 thk	0.5
14/1	LIFTING EYE	M.S.–IS2062	1	100x150x8 thk	0.9
14	LIFTING EYE WELDED TO TANK INCL. REINFORCEMENT PLATE.	M.S.	2		1.4X2=2.8
13/2	CHANNEL	M.S.–IS2062	1	ISMC 100X50–100 Lg	0.9
13/1	PAD PLATE	M.S.–IS2062	1	Sq.150x5 thk	0.92
13	BRACKETS WELDED TO TANK SHELL FOR HOLDING DOWN BOLTS.	M.S.	6		1.82x6=10.9
12/4	PLAIN WASHER	BRASS	2	ø40odxø18idx3 thk	--
12/3	EARTHING BOSS	M.S–IS2062	1	ø45X30 Lg	0.4
12/2	EARTHING CONDUCTOR STRIP	G.I.	1	F40x5 thk–3000 Lg	4.7
12/1	EARTHING PLATE	G.I.	1	Sq.300 x 5 thk	3.5
12	EARTHING BOSS WITH STUD & NUT WELDED TO TANK SHELL.	M.S.	2		6.6X2=13.2
11	TEE SCREWED TYPE FOR 50NB VENT PIPE CONNECTION.	M.S. CLASS "C"	1	50NBx50NBx50NB	0.9
10	ANGULAR CHECK VALVE SIZE 50 NB. (AS PER DRG. NO. CC.23) NOT IN FABRICATORS SCOPE OF SUPPLY	G.G. TYPE	2	50NB	--
9	BLIND FLANGE–100NB, 150#	M.S.–IS2062	1	ø230dx16thk	5.2
7	SUCTION PIPE ON TANK SHELL–50 NB	M.S.–IS1239	1	ø60.3odX4.5thkx2050Lg	12.7
6	BLIND FLANGE ON MANHOLE COVER–100 NB FOR ATG	M.S.–IS1239	1	ø60.3odX4.5thkx2285Lg	14.2
5/17	PIPE SPOOL–50NB	M.S–IS1239	1	ø60.3odx4.5thkx190 Lg	1.2
5/16	STOPPER RINGS	M.S	2	Sq.6 x200 Lg	0.1
5/15	RIVET	M.S	6	ø5 x10 Lg	--
5/14	GI COLLAR	G.I	1	20x250 Lgx2 thk	0.1
5/13	GI CHAIN	G.I	1	10gx150 Lg	--
5/12	BRASS RIVET	BRASS	2	ø5x32 Lg	--
5/11	DIP ROD	ALUMINIUM	1	Sq 12x2600 Lg	1.0
5/10	CAP FOR DIP ROD	ALUMINIUM	1	ø55x55 Lg	0.4
5/9	ADAPTOR	ALUMINIUM	1	ø80x80 Lg	1.3
5/8	DATUM PLATE	M.S–IS2062	2	200X200X6thk	3.8
5/7	RING WASHER	NEOPRENE RUBBER	1	ø75odix ø60idx3thk	--
5/6	EYE PIECE	ALUM. (CAST)	2	30x15x5thk	--
5/5	DIP CAP	ALUM. (CAST)	1	ø90dia.x70thk	1.4
5/4	GASKET PACKING	CAF	4	ø150odx ø60idx3thk	--
5/3	SORF FLANGE–50NB, 150#	M.S.–IS2062	3	ø150odx ø63idx16thk	6.7
5/1	DIP PIPE–50NB	M.S.–IS1239	1	ø60.3odX4.5thkx2250Lg	14.0
5	DIP / DRAIN PIPE WITH CAP,DIP ROD,ADAPTOR etc	M.S./ALUM	1	50NB	40.0
4/10	BEND 100NB–SHORT RADIUS	M.S.–IS1239	1	ø114.3odX5.4thk R=1.5d	3.5
4/9	SPLICED 100NB PIPE	M.S.–IS1239	1	ø114.3odX5.4thkx225Lg	3.2
4/8	BAFFLE PLATE	M.S–IS2062	1	5thkx110x155	0.7
4/7	STRAPS	M.S–IS2062	4	5thkx25WIDEx125Lg	0.5
4/6	FILL CAP	ALUM. (CAST)	1	HEX.ø90dia.x40thk	0.8
4/5	GASKET PACKING	CAF	2	ø230odx ø120idx3thk	--
4/4	STUDS & NUTS	M.S.–IS1367	34	M16X50Lg	6.8
4/3	SLIP–ON FLANGE–100NB, 150#	M.S.–IS2062	1	ø230odx ø117idx16thk	5.2
4/2	PAD FLANGE–100NB, 150#	M.S.–IS2062	2	ø230odx ø120idx16thk	10.4
4/1	FILL PIPE–100NB	M.S.–IS1239	1	ø114.3odX5.4thkx2130Lg	30.7
4	TANK LORRY DISCHARGE PIPE/FILL PIPE INCLUDING FILL CAP	M.S.–IS1239	1	100NB	61.8
3/5	GASKET PACKING	CAF	1	ø662odc ø520idx3thk	--
3/4	HEX.BOLTS & NUTS	M.S.–IS1367	24	M16X50Lg	4.8
3/3	MANHOLE NECK	M.S.–IS2062	1	165x165x5thk	10.7
3/2	MANHOLE FLANGE	M.S.–IS2062	1	ø662odx ø532idx8thk	21.6
3/1	MANHOLE COVER	M.S.–IS2062	1	ø662x10thk	27.0
3	MANHOLE COVER WITH NECK & COVER PLATE/PAD/STUDS/NUTS	M.S.–IS2062	1	ø520	64.1
2	TANK END PLATES 8 MM. THK.	M.S.–IS2062	2	ø2000	396.0
1	TANK SHELL PLATES 6 MM. THK.	M.S.–IS2062	3 1	1500x6300 500x6300	1483.7

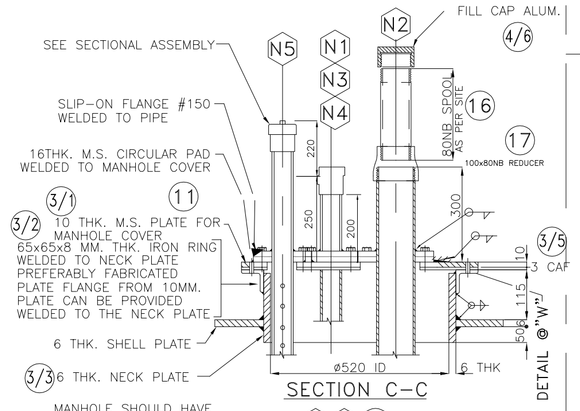


FILL PIPE CAP 4/6

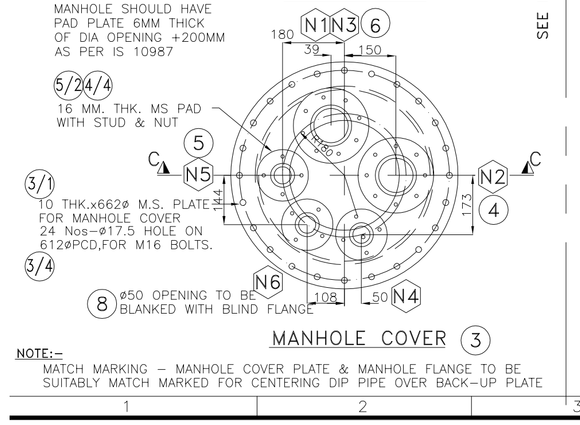


14 DETAIL OF LIFTING LUG



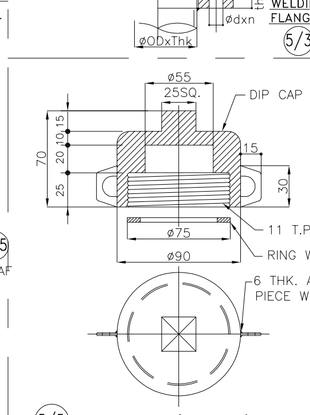


SECTION C–C

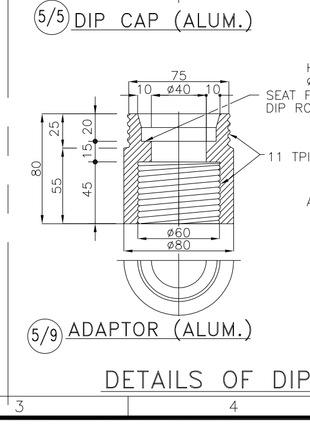


MANHOLE COVER 3

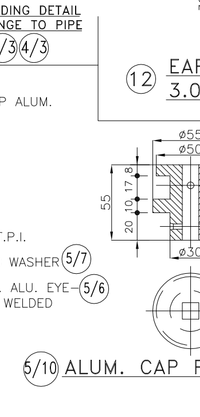
NOTE:–
MATCH MARKING – MANHOLE COVER PLATE & MANHOLE FLANGE TO BE SUITABLY MATCH MARKED FOR CENTERING DIP PIPE OVER BACK–UP PLATE



5/5 DIP CAP (ALUM.)



5/9 ADAPTOR (ALUM.)



5/10 ALUM. CAP FOR DIP ROD

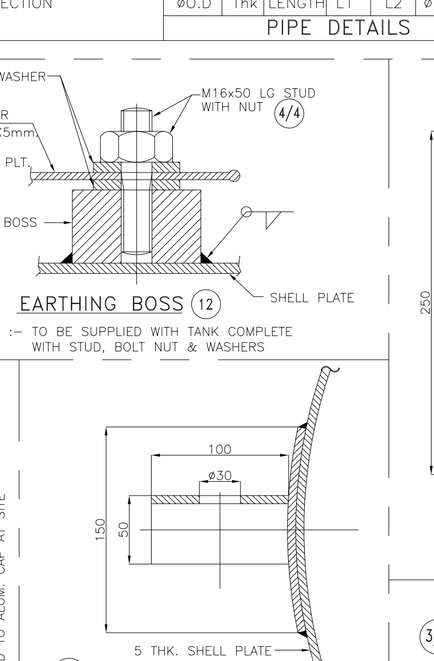
5/11 DIP ROD ALUM. 5 DIP ROD ASSEMBLY

DETAILS OF DIP/VENT/DRAIN PIPE FITTINGS

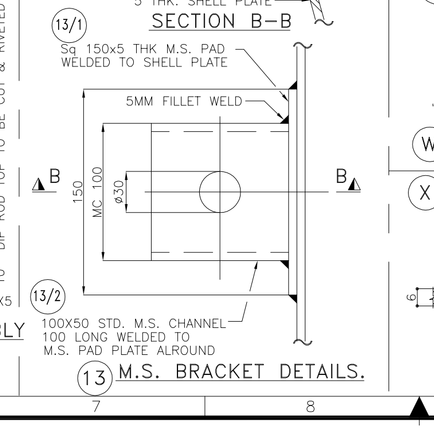
12 EARTHING (G.I. PLATE 3.0M BELOW EARTH)

EARTHING BOSS 12

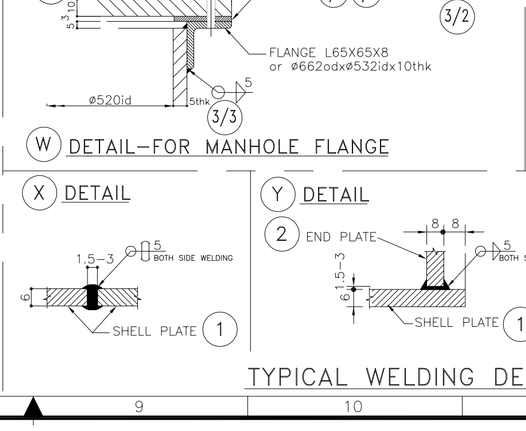
NOTE :– TO BE SUPPLIED WITH TANK COMPLETE WITH STUD, BOLT NUT & WASHERS



SECTIONAL ASSEMBLY 5



13 M.S. BRACKET DETAILS.



W DETAIL–FOR MANHOLE FLANGE

TYPICAL WELDING DETAILS

TANK DIMENSIONS

CAPACITY	15KL
INSIDE DIAMETER (mm)	2000
SHELL I/S LENGTH(mm)	4968
SHELL LENGTH (mm)	5000
TANK VOLUMETRIC CAP.	15.6cub.mtr
TANK VAPOUR CAPACITY	5%
SHELL PLATE DEV.(mm)	6300
END PLATE DIA. (mm)	2000

NOTES:

- DESIGN & CONSTRUCTION CODE : IS 10987 –1992
- TANK MATERIAL : ALL STEEL PLATES, STRUCTURAL STEEL TO CONFIRM TO I.S 2062
- TESTING : TANK AFTER FABRICATION WITH MANHOLE COVER & FITTINGS TO BE TESTED TO HYDRAULIC PRESSURE OF 0.50 KG. PER SQ. CM. WELD JOINTS TO BE HAMMER TESTED
- SHOP COAT : TANKS TO BE SCRAPED WITH WIRE BRUSH & CLEANED INTERNALLY & GIVEN A SHOP COAT OF RED LEAD PRIMER OF APPROVED QUALITY TO OUTER SURFACE.
- PIPES : SUCTION, VENT/DIP, FILL PIPES SHOULD BE M.S CLASS 'C' (IS1239)
- CRATING : ALUM. DIP ROD, FILL PIPE CAP FITTINGS, TO BE PACKED IN WOODEN BOX & LIST OF THESE ITEMS TO ACCOMPANY EACH TANK.
- "L" WELD SEAM : TO BE PROVIDED ON TOP HALF OF THE TANK.

NOTES:
ALL DIMENSIONS ARE IN MM.
DRAWING APPROVED BY C.C.O.E. NAGPUR VIDE LETTER NO. P–6 (5) 82 DTD 24–01–2006

भारत पेट्रोलियम कॉर्पोरेशन लिमिटेड
BHARAT PETROLEUM CORPORATION LTD.

SUBJECT : RETAIL OUTLET FACILITIES

FABRICATION DETAILS OF 15,000 LITRES CAPACITY CYLINDRICAL TANK FOR UNDER GROUND INSTALLATION

DATE | DRAWING NO.
NUMBERS CROSSED OUT SUPERCEDED DRWGS.
DRAWING NO : RE. DRG. 002

REVISION DRG. NO	DATE	ZONE OR PART	BRIEF RECORD	INITIALS
R1	30/07/04		SHELL THK., END STIFFNER ANGLE REVISED	

REVISIONS

SCALE : 1:25 | DRAWN: REDCO | TRACED | CHECKED
DATE : 20/12/99 | SR. MGR. (DESIGN & DEVP.)